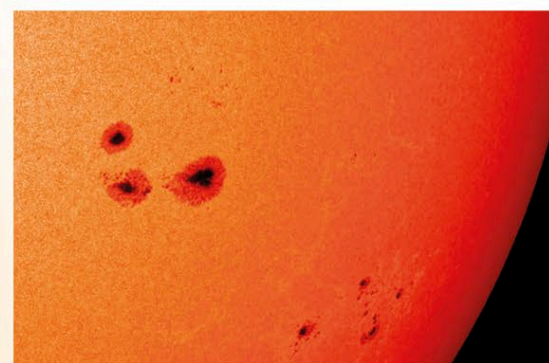


Prominence
eruption

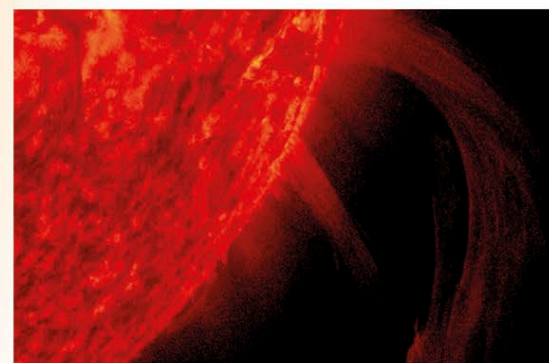
Surface features

Almost all the features seen on the Sun are controlled by its magnetic field. Large-scale dark patches in hot plasma indicate coronal holes, which are the source of much of the solar wind. The magnetic field at these holes escapes into space with the wind. Bright patches indicate tightly bunched, twisted magnetic fields trapping hot plasma, and are termed active regions. They usually overlie sunspots. When an active region erupts, it becomes a solar flare, which is bright at all wavelengths for minutes or longer.



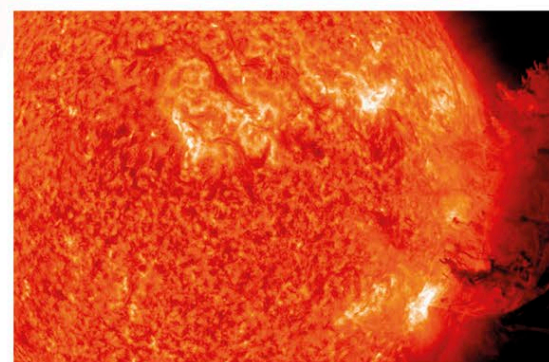
△ Sunspots

These dark regions are cooler than their surroundings, but are still extremely hot. They indicate where bundles of magnetic fields from the Sun's interior have broken through the surface.



△ Prominence

A lifted, plumelike region of cooler, denser plasma that emerges from the Sun's visible surface (the photosphere) is called a prominence.



△ Coronal mass ejections

These are eruptions of plasma that are sent out into space. They occur with a range of sizes and speeds. Some can have dramatic effects on Earth's magnetosphere.

This image is showing a layer in the Sun's atmosphere called the chromosphere, which lies above the visible photosphere

△ The Sun in ultraviolet light

By observing the Sun at different wavelengths, we see components of different temperature. This ultraviolet view shows material that is around 108,000°F (60,000°C), or ten times hotter than the visible surface. It is much more highly structured than the visible light view, showing bright areas of dense, hot plasma, and cooler, dark areas, termed filaments, suspended above the surface by magnetic fields.